

Mid-Term Presentation: Slide Explanations and Q&A

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Each section below matches one slide of the 12-slide mid-term deck: what the slide shows, and what to say in simple words. Plan about 1 minute per slide (10–12 minutes in total). Likely questions with short answers and the key numbers to remember are at the end.

Slide 1: Title

What it shows: project title, the one-line summary, your names, mentor and evaluator.

What to say:

Good morning. Our project is a multi-modal deep learning framework for early detection of skin diseases. In simple words, our model looks at two things together: the skin image, and basic patient data such as age, sex and the body site of the lesion. We use DenseNet121 for the image, add a small attention block that helps the model focus on the lesion, and combine it with the patient data. We test it on the HAM10000 dataset. This is a mid-term update, so we will show the design, what is done, and what comes next.

Tip: point to the small picture on the right: image + clinical data → diagnosis. That is the whole idea in one picture.

Slide 2: Introduction

What it shows: why skin cancer matters, and three numbers from the literature.

What to say:

Skin cancer is a major health problem worldwide. If it is found early, treatment is simpler, cheaper and more successful. Today, doctors diagnose it by looking at the skin, using the ABCDE rule and a dermoscope, which is a magnifying camera. This depends on the doctor's experience, and specialists are not available everywhere. Deep learning models trained on skin images can now perform as well as dermatologists. One study found a CNN reached an AUC of 0.86, against 0.79 for 58 dermatologists. But these models look only at the image and ignore the patient.

Words to know:

- **Dermoscopy:** taking a magnified, well-lit photo of a skin lesion.
- **ABCDE rule:** checklist for melanoma: Asymmetry, Border, Colour, Diameter, Evolving.
- **AUC:** area under the ROC curve; 1.0 is perfect, 0.5 is guessing.
- **CNN:** convolutional neural network, the standard deep learning model for images.

Slide 3: Problem & Motivation

What it shows: three problems on the left, three reasons for our approach on the right.

What to say (problems):

First, diagnosis is subjective: two doctors can judge the same lesion differently. Second, most AI models use only the image, but age, sex and body site also affect the diagnosis. For example, a lesion on an older person's face is judged differently from one on a young person's back. Third, real data is hard: some diseases have very few images, and doctors do not trust a model they cannot understand.

What to say (motivation):

So we combine the image with patient data, as a doctor does. Better automatic screening can help send patients to a specialist earlier. And we keep the model light and explainable: our attention block adds only 1,029 parameters, and heatmaps show where the model is looking.

Slide 4: Design & Methodology – Dataset

What it shows: the HAM10000 class chart on the left and the five data-preparation steps on the right.

What to say:

We use HAM10000, a public dataset of 10,015 skin lesion images in 7 classes, with age, sex and body site for each patient. The chart shows the main difficulty: NV, a normal mole, has 6,705 images, while DF has only 115. That is a 58 to 1 imbalance.

Then walk through the five steps:

1. **Lesion-level split.** Some lesions have several photos. We split by lesion, not by image, so photos of the same lesion never appear in both training and test: 70% training (7,002 images), 15% validation (1,532), 15% test (1,481).

2. **Metadata pre-processing.** Sex and body site become one-hot columns, and age is scaled. We remove dx type, the way the diagnosis was confirmed, because it almost gives away the answer.
3. **Class-balanced sampling.** In training only, each epoch takes 2,000 images from every class (14,000 in total), so rare classes are seen more often. Validation and test keep the real class balance.
4. **Image pre-processing.** Images are resized to 224×224 and normalised with ImageNet values. During training we also flip, rotate, zoom and slightly change the colours, so the model sees varied copies.
5. **Modality alignment.** Each image is matched to its patient record through image_id.

The note at the bottom:

The reference paper's test set seems to contain repeated images. Its confusion matrix shows 435 DF test samples, but the whole dataset has only 115 DF images. This is why we avoid that setup, and why our numbers may be lower but more honest.

Words to know:

- **One-hot encoding:** turning a category (for example "back") into a column of 0s and 1s.
- **Epoch:** one full pass of training over the training data.
- **Data leakage:** when test information sneaks into training, making results look better than they are.

Slide 5: Design & Methodology – Proposed Architecture

What it shows: the full model as a block diagram, with the attention block in orange as our contribution.

What to say (follow the arrows left to right):

The model has two branches. The top branch handles the image. The image is resized and normalised, then DenseNet121 turns it into a feature map of size $7 \times 7 \times 1024$. You can think of this as 7×7 small regions of the image, each described by 1024 numbers. DenseNet121 is pretrained on ImageNet, so it already knows general shapes and textures.

Next is our attention block, shown in orange. It gives more weight to the regions with the lesion and less to the background, such as normal skin or hair. Then global average pooling, GAP, averages the 7×7 regions into a single list of 1024 numbers.

The bottom branch handles the patient data. Age, sex and body site go through a small MLP, which turns them into 64 numbers.

Finally, we join the two lists, 1024 plus 64 gives 1088 numbers. A dense layer and a final FC layer with softmax then give a probability for each of the 7 classes.

Why these choices (if asked):

- **DenseNet121:** strong on medical images, fewer parameters than ResNet50, and fits a 6 GB GPU.
- **Concatenation:** the simplest way to combine the two branches, so any gain clearly comes from the attention block and the patient data.
- **Softmax:** turns the 7 outputs into probabilities that add up to 1.

Slide 6: Design & Methodology – Spatial Attention Module

What it shows: the inside of our attention block. This is the main contribution, so spend a little more time here (about 2 minutes).

What to say (follow the diagram):

The input is the DenseNet feature map F , of size $7 \times 7 \times 1024$. The block's job is to learn a map M of size 7×7 with one weight per region, saying how important that region is.

F goes through three paths. Each path squeezes the 1024 channels into one channel, giving a $7 \times 7 \times 1$ map.

Path 1 is a 1×1 convolution. It is learned during training, so the model decides for itself which features matter.

Path 2 is max pooling across channels. For each region it keeps the strongest response, which highlights sharp, stand-out features such as borders.

Path 3 is average pooling across channels, called A , which shows the overall activity of each region. We then reshape A into one row of 49 values, apply a sigmoid, and multiply the result with A . This self-gating keeps strong regions and weakens weak ones.

The three maps are joined into $7 \times 7 \times 3$. A second 1×1 convolution mixes them into one map, and a sigmoid squeezes the values between 0 and 1. This is the attention map M .

Finally, F is multiplied by M at every position, through the identity path at the bottom. Regions with a weight near 1, such as the lesion, are kept; regions with a weight near 0, such as background, are reduced. The output has the same size as the input, so the block can be placed anywhere.

The whole block has only 1,029 trainable parameters: 1,025 in the first convolution and 4 in the second. Pooling, reshape and sigmoid have none.

Simple analogy: the block works like a highlighter pen. It marks which parts of the image the classifier should read carefully.

Words to know:

- **1×1 convolution:** a learned weighted sum across channels at each position; it does not look at neighbouring positions.
- **Sigmoid:** a function that maps any number into the range 0 to 1.
- **Channel:** one of the 1024 feature types DenseNet computes for each region.

Slide 7: Design & Methodology – Experimental Design

What it shows: how we will prove that each part of the model helps.

What to say:

We will not just train one model. We plan four groups of experiments.

1. **Ablation study (2×2).** We start with DenseNet121 on images only. Then we add the attention block, then add the patient data, then both. If each step improves the result, we know each part is useful.
2. **Comparison of attention modules.** We keep everything the same and only swap the attention block: no attention, SE, CBAM, and ours. SE and CBAM are two well-known attention blocks, each adding about 132,000 parameters, while ours adds 1,029.
3. **Statistical robustness.** Results can change a little with random starting weights. We train the key models three times with different random seeds and report the average and spread. McNemar's test then checks whether our model's improvement is real or just luck.
4. **Evaluation metrics.** Because the data is imbalanced, plain accuracy is misleading. We report balanced accuracy, macro F1 and macro AUC, melanoma sensitivity and specificity, ROC curves, and heatmaps from the attention block and Grad-CAM.

Words to know:

- **Ablation study:** removing or adding one part at a time to see what it contributes.
- **Seed:** the starting value of the random number generator; a different seed gives a slightly different training run.
- **Sensitivity:** of all real melanomas, how many the model finds. **Specificity:** of all non-melanomas, how many it correctly clears.
- **Grad-CAM:** a heatmap showing which image regions drove a prediction.

Slide 8: Implementation & Technical Details

What it shows: the main settings in four boxes.

What to say:

Image branch: DenseNet121 pretrained on ImageNet, followed by our attention block and global average pooling. BatchNorm layers are kept fixed during fine-tuning, because our batches are small.

Clinical branch: standardised age plus one-hot sex and body site, passed through two layers with 128 and 64 units. We removed dx type because it almost reveals the label.

Training: we train in two stages. In stage 1, DenseNet is frozen and only the new layers learn, for 5 epochs with a learning rate of 0.001. In stage 2, we unfreeze the last 150 DenseNet layers and train with a smaller learning rate of 0.0001, so the pretrained knowledge is not destroyed. We also use label smoothing, early stopping and mixed precision.

Software and hardware: TensorFlow 2.10 with Keras, on an NVIDIA RTX 3050 laptop GPU with 6 GB of memory, batch size 16.

Words to know:

- **Fine-tuning:** continuing to train a pretrained network on our own data.
- **Label smoothing:** makes the model slightly less over-confident.
- **Early stopping:** stops training when validation loss stops improving, to avoid overfitting.
- **Mixed precision:** uses 16-bit numbers where safe, which saves GPU memory and time.

Slide 9: Results (Preliminary)

What it shows: the only result so far: a model trained on patient data alone (no images), tested on 1,481 test images.

What to say:

This is a baseline, not our final model. Using only age, sex and body site, the model reaches 62.46% accuracy, with macro precision 0.32, macro recall 0.39 and macro F1 0.33.

Macro scores are much lower than accuracy. That tells us the model handles the common class, NV, but often misses the rare classes. About 67% of the test images are NV, so a model that always says NV would already get about 67% accuracy while being useless. That is why we report macro scores, which treat all 7 classes equally.

The reference paper reports 77% for its clinical-only model, but it is not a fair comparison because we split by lesion and keep the test set untouched. The main message: patient data alone is weak, so we need the images. Training of the image and combined models is in progress.

Words to know:

- **Macro average:** compute the score for each class, then take the plain average, so small classes count as much as big ones.
- **Precision:** of the cases predicted as a class, how many were right. **Recall:** of the real cases of a class, how many were found.

Slide 10: Conclusion & Future Work

What it shows: what is done, what is running, what is planned, and the next four steps.

What to say:

Completed: the data pipeline, the clinical-only baseline, and the full TensorFlow code for the proposed model and all comparison models.

In progress: training the proposed model and the comparison models, including SE and CBAM.

Next steps: first, train all models and report the ablation and attention comparison. Second, repeat the key models with three seeds and check significance with McNemar's test. Third, explain predictions with attention maps and Grad-CAM, and report how well we detect melanoma. Fourth, test on a second dataset, ISIC 2019 or PAD-UFES-20, and write the research paper.

To conclude: patient data alone is not enough. Combining it with image features refined by our attention block is the way forward.

Slides 11–12: Q&A and Thank you

What to say on slide 11:

Thank you for listening. We are happy to take questions. Some points we can discuss: why we split by lesion, how our attention block compares with SE and CBAM, and our next experiments.

Slide 12: thank the mentor and the evaluator by name. No need to say more.

Likely questions and simple answers

About the idea

- 1. What is new in your work?** A small spatial attention block, designed with our mentor, placed after DenseNet121 and combined with patient data. It has only 1,029 parameters, against about 132,000 for SE or CBAM. We also use a fair, lesion-level split.
- 2. Why use patient data at all? Isn't the image enough?** Doctors use age and body site too. For example, a lesion on the face of an older person is more likely to be a carcinoma. The image is the main signal, but patient data adds context that helps on hard cases.
- 3. Why DenseNet121 and not ResNet or EfficientNet?** DenseNet121 reuses features well, works well on medical images, has about 8 million parameters (ResNet50 has about 25 million), and fits our 6 GB GPU. We can test other backbones later; the block works with any of them.

About the attention block

- 4. What does the attention block actually do?** It learns one weight between 0 and 1 for each of the 7×7 regions of the feature map and multiplies the features by it. The lesion area stays, the background is reduced.
- 5. How is it different from CBAM?** CBAM's spatial part uses average and max pooling with a 7×7 convolution, and CBAM also adds a channel-attention part. Ours adds a learned 1×1 convolution path and a self-gated average path, and has no channel part. That makes it about 128 times smaller in parameters. Our experiments will show whether it is also as accurate.
- 6. Why reshape to $1 \times HW$ before the sigmoid?** This follows our mentor's design. The sigmoid works on each value separately, so the reshape does not change the result; it shows that the gate is computed over all spatial positions. A softmax over the positions could be tried as a variant.

7. Why is the map applied at 7×7 ? Isn't that coarse? Yes, each cell covers about 32×32 pixels. It is enough to separate the lesion from the background, and it keeps the block cheap. Using it at an earlier, larger layer is possible future work.

About data and evaluation

8. Why split by lesion and not by image? Some lesions have several photos. If one photo is in training and another in test, the model has effectively seen the test case, and the score is too high. Splitting by lesion prevents this.

9. Why did you remove dx type? It records how the diagnosis was confirmed, for example by biopsy. It is known only after diagnosis and almost reveals the class, so using it would be cheating.

10. How do you handle class imbalance? During training we sample the same number of images from each class every epoch, with random augmentation, so rare classes are seen more often. The test set keeps the real imbalance, and we report macro and balanced scores.

11. Why is your clinical-only accuracy (62%) lower than the paper's (77%)? We use a lesion-level split and an untouched test set, and the paper's test set appears to contain repeated images. Our training is also balanced, which trades some accuracy for better detection of rare classes.

12. Why not just report accuracy? About 67% of test images are NV. A model that always says NV gets about 67% accuracy and is useless. Macro F1 and balanced accuracy treat every class equally.

13. Why is melanoma sensitivity important? Missing a melanoma is the most dangerous error, because it can spread. A false alarm only leads to an extra check.

About results and next steps

14. Where are the results of the full model? The code is complete and tested, and the models are training now. Each DenseNet run takes several hours on our laptop GPU, so the full results will be in the final evaluation.

15. How will you know the attention block really helps? By the ablation (with and without the block), the comparison with SE and CBAM, three seeds per key model, and McNemar's test for significance. The attention maps will also show whether it looks at the lesion.

16. Can this be used in hospitals? Not yet. It is a research prototype. It would need testing on other datasets and populations, and approval, before clinical use. It is meant to support doctors, not replace them.

17. Why TensorFlow and not PyTorch? We started in PyTorch, then moved to TensorFlow/Keras for this architecture because its pretrained DenseNet121 and training tools made the two-stage fine-tuning simple.

18. What are the limitations? HAM10000 comes mostly from fair-skinned European patients, has few images for some classes, and has only three patient features. The model may perform worse on other skin types and cameras, which is why we plan external testing.

Key numbers to remember

What	Number
Images in HAM10000	10,015
Classes	7 (MEL, NV, BCC, AKIEC, BKL, DF, VASC)
Largest vs smallest class	NV 6,705 vs DF 115 (58:1)
Train / validation / test	7,002 / 1,532 / 1,481 (70 / 15 / 15%)
Image size	224 × 224
DenseNet121 output	7 × 7 × 1024
Image vector after GAP	1024
Clinical vector after MLP	64 (layers 128 → 64)
Joined vector	1088
Attention block parameters	1,029 (SE or CBAM: about 132,000)
Training per epoch	2,000 images per class, 14,000 in total
Stage 1 / stage 2 learning rate	0.001 for 5 epochs / 0.0001 for up to 30 epochs
Clinical-only baseline	62.46% accuracy, 0.33 macro F1
Share of NV in test set	about 67%